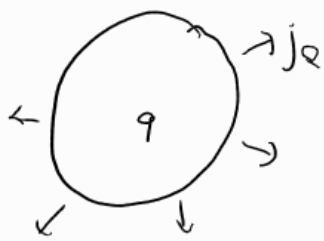


(5.2) Kontinuitetna enačba v Zemlji:



q ... gostota virov
toplote

$j_Q = \frac{\frac{4}{3}\pi r^3 q}{4\pi r^2} = \frac{1}{3} r q \rightarrow \vec{j}_Q = \frac{1}{3} r \hat{e}_r = \frac{1}{3} q(x, y, z)$

← vsi viri
površina sfere

Pokaži, da velja $\nabla \cdot \vec{j}_Q = q$

$$\nabla \cdot \vec{j}_Q = \frac{\partial j_x}{\partial x} + \frac{\partial j_y}{\partial y} + \frac{\partial j_z}{\partial z} = \frac{1}{3} q + \frac{1}{3} q + \frac{1}{3} q = q \checkmark$$

(5.3) Brez pisanja komponent pokaži, da je $\nabla \cdot \frac{\vec{r}}{r} = \frac{2}{r}$



$$\int \nabla \cdot \frac{\vec{r}}{r} dV = \oint \frac{\vec{r}}{r} d\vec{S} = \oint \frac{r}{r} dS = 4\pi R^2$$

$$\int_0^R 4\pi r^2 dr \nabla \cdot \frac{\vec{r}}{r} = 4\pi R^2$$

Če želimo, da je to enako, mora $\nabla \cdot \frac{\vec{r}}{r} = \frac{2}{r}$

(5.9) 74/3

silna tokarnega vodila na točkast magn. dipol



$$\vec{F} = (\vec{p} \cdot \nabla) \vec{B}$$

$$\hat{e}_\varphi = -\sin\varphi \hat{i} + \cos\varphi \hat{j}$$

$$\frac{\partial \hat{e}_\varphi}{\partial \varphi} = -\cos\varphi \hat{i} - \sin\varphi \hat{j} = -\hat{e}_r$$

$$\vec{B} = \frac{\mu_0 I}{2\pi r} \hat{e}_\varphi \quad \vec{p} = p \cos\theta \hat{e}_r + p \sin\theta \hat{e}_\varphi$$

$$\begin{aligned} \vec{F} &= p((\cos\theta \hat{e}_r + \sin\theta \hat{e}_\varphi) \cdot \nabla) \frac{\mu_0 I}{2\pi r} \hat{e}_\varphi = \frac{p\mu_0 I}{2\pi} \left(\cos\theta \frac{\partial}{\partial r} + \sin\theta \frac{1}{r} \frac{\partial}{\partial \varphi} \right) \frac{1}{r} \hat{e}_\varphi = \\ &= \frac{p\mu_0 I}{2\pi} \left(\cos\theta \left(-\frac{1}{r^2} \hat{e}_\varphi \right) + \sin\theta \frac{1}{r} \cdot \frac{1}{r} (-\hat{e}_r) \right) = -\frac{p\mu_0 I}{2\pi r^2} (\sin\theta \hat{e}_r + \cos\theta \hat{e}_\varphi) \end{aligned}$$

